

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ403			
Course Title	RINGS AND MODULES			
Type of Course	Elective			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Elementary number theory, algebra, combinatorics, basic linear algebra			
Course Summary	This course is a self-contained elementary introduction to Rings and Modules. The course will cover basic topics of Ring Theory and Module Theory which is a core course in Algebra			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and differentiate between various types of rings, including rings of continuous functions, matrix rings and polynomial rings	U	C	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO2	Analyse and apply the concepts of ideals within rings, including definitions, maximal ideals, generators for subrings and ideals.	An	Ap	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
CO3	Evaluate and synthesize the concepts of homomorphisms of rings, including quotient rings, ideals in quotient rings, endomorphism rings and field of fractions.	E	M	Internal Exam/Assignment/Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text book Introduction to Rings and Modules, C. Musili, Narosa Publishing House, 2001.				
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)
I	Rings		12	
	1	Chapter 1 – Section 1.1: Terminology		
	2	Chapter 1 – Section 1.2: Rings of Continuous functions		
	3	Chapter 1 – Section 1.3 to 1.5: Matrix Rings, Polynomial Rings and Power series rings		
	4	Chapter 1 – Section 1.8 to 1.9: Some Special Rings and Direct Products		
	5	Chapter 1 – Section 1.10 to 1.12: Several Variables, Opposite rings, Characteristic of a ring		
II	Ideals		12	
	6	Chapter 2 – Section 2.1 to 2.2 : Definitions, Maximal Ideals		
	7	Chapter 2 – Section 2.3: Generators for subrings and Ideals		
	8	Chapter 2 – Section 2.4: Basic Properties of Ideals		
	9	Chapter 2 – Section 2.5: Algebra of Ideals		
III	Homomorphisms of Rings		12	
	10	Chapter 2 – Section 2.6 & 2.7 : Quotient rings and Ideals in Quotient rings		
	11	Chapter 3 – Section 3.1: Definition and Basic Properties		
	12	Chapter 3 – Section 3.2 : Fundamental Theorems of Homomorphisms		
	13	Chapter 3 – Section 3.3: Endomorphism Rings		
	14	Chapter 3 – Section 3.4: Field of Fractions		
	15	Chapter 3 – Section 3.5: Prime Fields		
IV	Modules		12	
	16	Chapter 5: Modules: Section 5.1: Definition and Examples		
	17	Chapter 5: Section 5.2 to 5.4: Direct sums, Free Modules and Vector spaces		
	18	Chapter 5: Section 5.4 to 5.3: Direct sums and Free Modules		
	19	Chapter 5: Section 5.6: Quotient Modules		
	20	Chapter 5: Section 5.7: Homomorphisms		
	21	Chapter 5: Section 5.8: Simple Modules		
V	Open Ended		12	
	Artinian Modules and Rings, Noetherian Modules and Rings, Nil Radical, Jacobson Radical			
References	1. John B. Fraleigh, A First Course in Abstract Algebra, 7th Edition, 2002 2. M. Artin: Algebra, Prentice Hall, 1991 3. Thomas W. Hungerford, Algebra, Springer, 2003 4. Joseph Gallian, Contemporary Abstract Algebra, 7th Edition, Cengage Learning, 2009. 5. D.M. Burton, A First Course in rings and ideals, Addison- Wesley, 1970.			

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	1	2	1	3	0	1
CO 2	2	3	2	1	3	1	3	1	3	0	1
CO 3	2	2	2	1	3	1	3	1	3	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓